Worksheet No. - 2

Student Name: Anjali kumari
UID: 25MCA20023
Branch: MCA general
Section: Group:1/A
Semester:3rd
Date of Performance:12th Aug.

Subject Name:MACHINE LEARNING.

Subject Code: CONT_25CAP-702

Experiment 1: Write a program to implement Linear regression on Jupyternotebook.

a. Download different datasets from Kaggle or UCI MLrepository.
b. Import all the necessary modules for the datasets.
c. Find out the best fit line along with the MSE, RMSE.

1.Aim:

To implement Linear Regression on the **Student Performance dataset** using Jupyter Notebook and predict students' performance based on relevant academic and other factors. The model will be evaluated using **Mean Squared Error (MSE)** and **Root Mean Squared Error (RMSE)**, and the best-fit regression line will be visualized.

2.Objective:

- Load and explore the Student Performance dataset.
- Preprocess the data and select suitable features.
- Train a Linear Regression model using training data.
- Find the best-fit regression line and make predictions.
- Evaluate the model using **MSE and RMSE**.

3.Task to be done :

- Understand the concept and implementation of Linear Regression.
- Learn to preprocess and analyze a real-world student dataset.
- Develop a regression model using Python and Scikit-learn.
- Learn to visualize the best-fit line and predicted values.
- Evaluate regression performance using MSE and RMSE.

4.Online Resources:

1. NumPy Documentation
2. Pandas Documentation
3. Scikit-Learn Official Documentation
4. Matplotlib Official Documentation
5. Seaborn Documentation
6. Kaggle Titanic Dataset Resources

Software Requirements

1. Python 3.x
2. Jupyter Notebook / Google Colab
3. Anaconda Distribution

Standard Tools Used

1. NumPy
2. Pandas
3. Matplotlib
4. Seaborn
5. SciPy
6. Scikit-Learn

5.Procedure/ Algorithm:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

# Load dataset
df = pd.read_csv(r"C:\Users\ASUS\Downloads\Student_Performance.csv")

# Display dataset
print("Dataset:")
print(df.tail())

# Select independent and dependent variables
X = df[['study_hours']]
y = df['overall_score']

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

# Create Linear Regression model
model = LinearRegression()
```

```
# Train model
model.fit(X_train, y_train)

# Find slope and intercept
m = model.coef_[0]
c = model.intercept_

print("\nSlope:", m)
print("Intercept:", c)

# Best fit line
print(f"\nBest Fit Line:")
print(f"Overall Score = {m:.2f} × Study Hours + {c:.2f}")

# Fill numerical columns with mean
for col in numerical_cols:
    df[col] = df[col].fillna(df[col].mean
print("\nMissing Values After Filling:\n")
print(df.isnull().sum())


# 3. Label Encoding

encoder = LabelEncoder()

for col in categorical_cols:
    df[col] = encoder.fit_transform(df[col])

print("\nFirst 5 Rows After Encoding:\n")
print(df.head())


# 5. Standardize Numerical Values

standard_scaler = StandardScaler()

if len(numerical_cols) > 0:
    df[numerical_cols] = standard_scaler.fit_transform(df[numerical_cols])

print("\nStandardized Numerical Columns:\n")
print(df[numerical_cols].head())


# Final Dataset

print("\nFinal Dataset Preview:\n")
print(df.head()) …………………………………………………………………continue,
```

6.output:

```
Dataset:
       student_id  age  gender school_type parent_education  study_hours  \
24995       12047   17  female      public              phd          1.8
24996        1102   16  female     private          diploma          2.7
24997        4422   19   other     private    post graduate          1.0
24998        7858   14    male     private          diploma          1.0
24999       11621   18   other      public        no formal          0.7

       attendance_percentage internet_access travel_time extra_activities  \
24995                   55.2             yes   15-30 min               no
24996                   97.1             yes     <15 min               no
24997                   63.0             yes     <15 min               no
24998                   69.4             yes   15-30 min              yes
24999                   60.3             yes   30-60 min               no

      study_method  math_score  science_score  english_score  overall_score  \
24995        mixed        55.8           48.5           46.7           46.1
24996     coaching        64.8           48.2           52.3           56.5
24997  group study        50.5           20.3           36.1           36.7
24998  group study        13.0           34.2            7.3           34.1
24999        mixed        36.5           45.1           16.5           31.4

      final_grade
24995           e
24996           d
24997           f
24998           f
24999           f

Slope: 7.8987757672622845
Intercept: 30.39321599143166

Best Fit Line:
Overall Score = 7.90 × Study Hours + 30.39

MSE: 65.38461432368372
RMSE: 8.086075335024015
```

Fig. 1: Student Performance Dataset

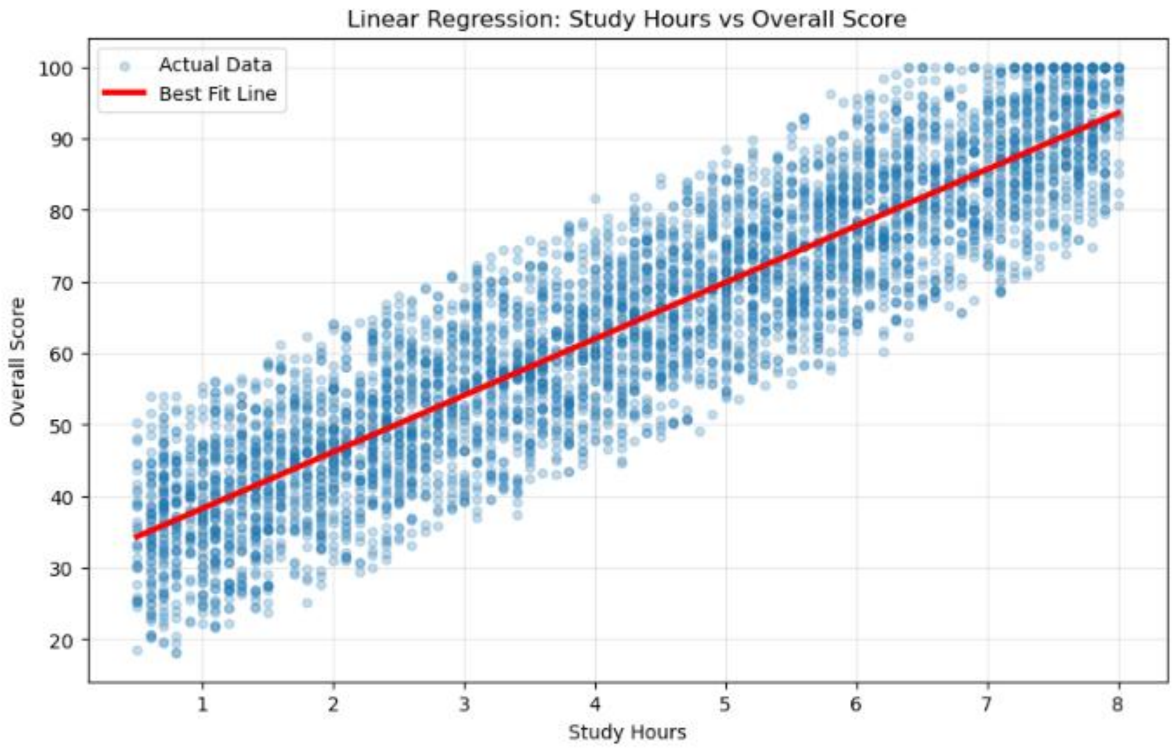

Fig. 2: Best-Fit Regression Line for Student Performance

6.Learning outcomes (what I have learnt) :

After completing this practical, you will be able to:

1. Understand the concept and working of Linear Regression.
2. Apply Linear Regression to a real-world Student Performance dataset.
3. Perform basic data preprocessing and exploratory data analysis using Python.
4. Divide data into training and testing datasets for machine learning.
5. Build and train a Linear Regression model using Scikit-learn.
6. Visualize the best-fit regression line and predicted values.
7. Calculate and interpret MSE and RMSE for model evaluation.
8. Analyze how different student-related factors can influence academic performance.

.